

NACOGDOCHES, TX, USA

WMO: 722499

Lat: 31.578N

Lon: 94.709W

Elev: 108

StdP: 100.03

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 53948

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-3.9	-2.5	-11.2	1.5	0.1	-8.0	1.9	2.9	9.8	11.4	8.7	11.6	1.2	350	0.389

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	12.2	37.3	24.3	36.0	24.5	34.0	24.4	26.4	32.2	25.8	31.8	25.5	31.5	3.1	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.1	20.4	28.2	24.2	19.3	27.7	23.9	19.1	27.7	82.4	32.0	80.2	32.2	78.9	31.3	28.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8.2	7.3	6.2	-7.9	38.2	2.4	2.1	-9.7	39.7	-11.1	40.9	-12.4	42.1	-14.1	43.6
			-8.5	27.2	2.2	0.7	-10.1	27.7	-11.4	28.1	-12.6	28.5	-14.2	28.9
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	18.9	8.9	11.2	14.7	18.7	23.1	26.5	28.0	28.1	24.9	19.3	13.8	9.6
	DBStd	8.09	5.58	5.67	5.16	4.26	3.34	2.24	1.85	2.10	3.04	4.41	5.10	5.36
	HDD10.0	250	89	48	17	1	0	0	0	0	0	1	20	74
	HDD18.3	1163	296	206	134	47	6	0	0	0	2	43	153	276
	CDD10.0	3518	55	83	163	262	407	496	557	562	448	288	133	63
	CDD18.3	1387	3	8	22	58	154	246	299	304	199	72	16	6
	CDH23.3	13876	9	36	139	462	1354	2497	3280	3452	1904	637	90	16
CDH26.7	6372	0	3	17	94	487	1164	1686	1848	876	187	9	1	
Wind	WSAvg	2.5	2.8	2.9	3.1	3.2	2.8	2.4	2.0	1.9	2.2	2.4	2.6	
Precipitation	PrecAvg	1260	117	113	110	106	109	109	75	88	95	109	118	126
	PrecMax	1713	292	234	232	264	292	316	199	258	216	360	389	279
	PrecMin	653	17	6	35	28	2	17	12	0	45	0	6	31
	PrecStd	278	74	65	51	59	68	70	49	70	48	99	89	60
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	24.1	27.1	28.1	31.3	33.9	37.4	38.0	39.2	37.3	32.3	27.8	25.9
		MCWB	17.5	18.2	19.6	22.0	23.4	24.3	24.3	24.4	23.1	23.1	21.4	20.2
	2%	DB	22.5	24.0	27.1	28.9	32.4	35.2	37.1	37.4	34.1	30.9	26.1	22.8
		MCWB	16.9	17.9	19.1	21.1	23.5	24.6	24.5	24.4	23.4	22.9	19.9	18.7
	5%	DB	20.9	22.4	24.9	27.6	31.1	33.5	35.1	36.1	32.8	28.8	23.8	21.3
		MCWB	16.6	17.3	17.9	20.1	23.2	24.5	24.7	24.5	23.3	21.7	18.7	18.4
	10%	DB	18.7	20.9	22.8	26.3	29.1	32.4	33.7	34.1	32.0	27.3	22.4	19.0
		MCWB	15.8	16.6	17.2	19.1	22.6	24.5	24.8	24.5	23.1	20.7	18.4	16.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	19.7	20.6	21.5	23.6	25.7	26.9	26.9	26.9	26.2	25.4	22.8	21.3
		MCDB	22.7	23.6	25.9	28.3	30.9	31.8	32.2	33.0	31.3	29.6	26.4	23.5
	2%	WB	18.7	19.6	20.5	22.5	24.8	26.0	26.3	26.3	25.5	24.4	21.6	20.1
		MCDB	21.1	22.6	24.2	26.8	30.0	31.3	32.1	32.8	30.8	28.5	24.2	22.2
	5%	WB	17.7	18.6	19.6	21.7	24.1	25.5	25.8	25.8	24.9	23.4	20.5	19.0
		MCDB	19.4	21.5	23.0	25.9	28.9	31.1	31.8	32.2	30.0	26.9	23.0	20.8
	10%	WB	15.9	17.2	18.8	20.9	23.4	25.1	25.4	25.4	24.3	22.3	19.0	17.0
		MCDB	18.3	19.5	22.4	24.3	28.0	30.6	31.5	31.8	29.0	25.3	21.8	18.7
Mean Daily Temperature Range	5% DB	MDBR	12.1	11.9	12.4	12.7	11.4	11.0	11.4	12.2	12.3	12.9	12.5	12.1
		MCDBR	13.8	13.3	13.1	13.0	12.2	12.5	13.4	14.1	14.1	13.4	13.3	13.0
		MCWBR	8.9	8.1	7.1	6.5	5.1	4.4	3.9	4.0	5.2	6.6	8.1	9.3
	5% WB	MCDBR	10.3	10.5	10.6	10.1	10.5	10.5	11.3	11.9	11.2	10.6	10.9	10.5
		MCWBR	8.0	7.6	7.0	5.8	5.1	4.4	4.1	4.0	4.6	6.0	7.9	8.6
Clear-Sky Solar Irradiance	taub	0.325	0.344	0.362	0.407	0.443	0.462	0.477	0.472	0.430	0.368	0.357	0.332	
	taud	2.522	2.477	2.427	2.307	2.250	2.244	2.234	2.243	2.319	2.474	2.455	2.511	
	Ebn at Noon	903	916	919	885	851	831	817	817	840	877	855	873	
	Edh at Noon	86	99	111	130	139	140	140	138	123	98	92	83	
All-Sky Solar Radiation	RadAvg	2.78	3.10	4.34	5.36	5.86	6.25	6.31	5.96	5.07	4.16	3.05	2.41	
	RadStd	0.25	0.54	0.41	0.34	0.43	0.47	0.46	0.45	0.47	0.57	0.35	0.30	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (31 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+85

Nomenclature: See separate page